FOR OFFICIAL USE

N5

National Qualifications 2026

Mark

X819/75/01 Design and Manufacture

WEDNESDAY, 13 MAY
1:00 PM – 2:45 PM

X8197501

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day	Month	Year

Scottish candidate number

Total marks — 80

SECTION 1 — 60 marks

Attempt ALL questions.

SECTION 2 — 20 marks

Attempt ALL questions.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Do not remove any exam materials. You must leave this booklet on your desk; if you do not, you could lose all the marks for this paper.

Qualifications Scotland
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SECTION 1 — 60 marks

Attempt ALL questions

1. A design proposal for a coffee station is shown below.



(a) A softwood was considered for the back, base and side.

(i) State **two** benefits of using softwoods in the manufacture of the coffee station. **2**

Benefit 1 _____

_____

_____

_____

Benefit 2 _____

_____

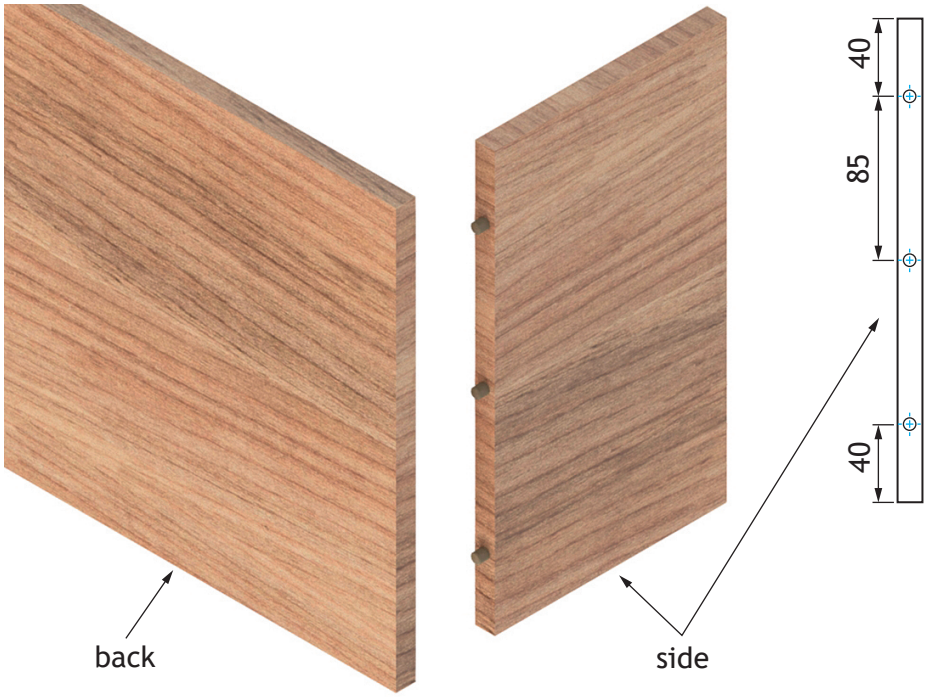
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1. (a) (continued)

The back and side of the coffee station were assembled using a dowel joint.

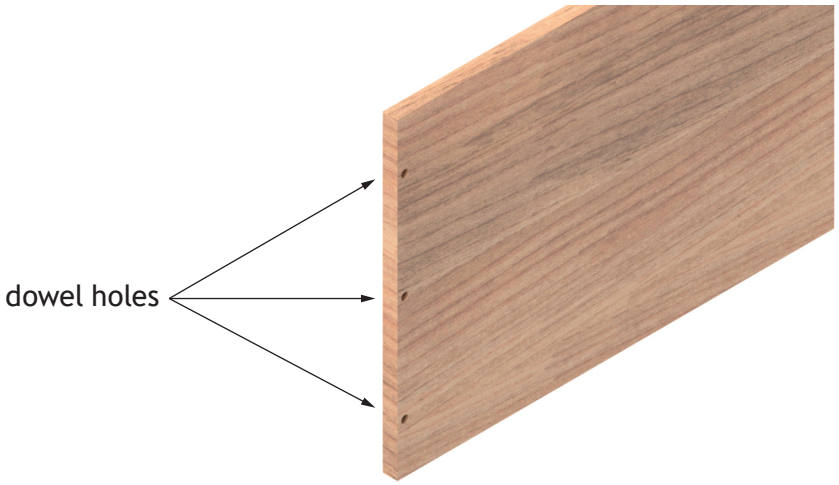


(ii) Describe how to **mark out** the position of the dowels on the side, with reference to workshop tools. **2**

You may use sketches to illustrate your answer in the box below.

1. (a) **(continued)**

The dowel holes in the back were drilled to a depth of 6 mm.



(iii) Name a suitable drill bit that could be used to make the holes. **1**

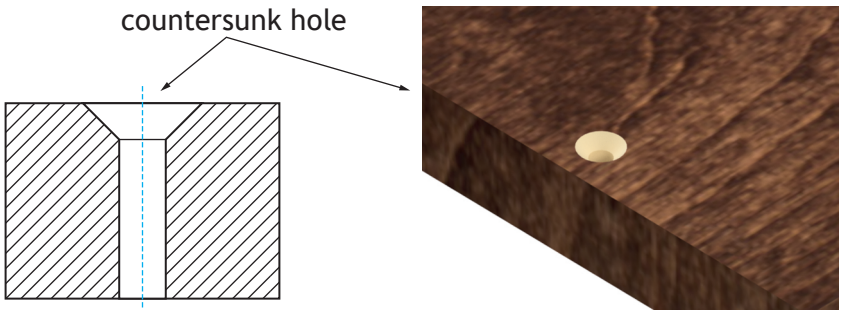
(iv) Outline one way of ensuring the holes are drilled to the correct depth. **1**

All wooden parts were prepared before a finish was applied.

(v) Describe **three** stages in the preparation of the wood before applying a finish. **3**

1. (a) **(continued)**

The base was stained then joined to the side and back using screws in countersunk holes.



(vi) State **two** reasons why wood stain is a suitable finish for the base. **2**

Reason 1 _____

_____

_____

Reason 2 _____

_____

_____

The holes on the underside of the base were countersunk.

(vii) Explain why the holes were countersunk. **1**

_____

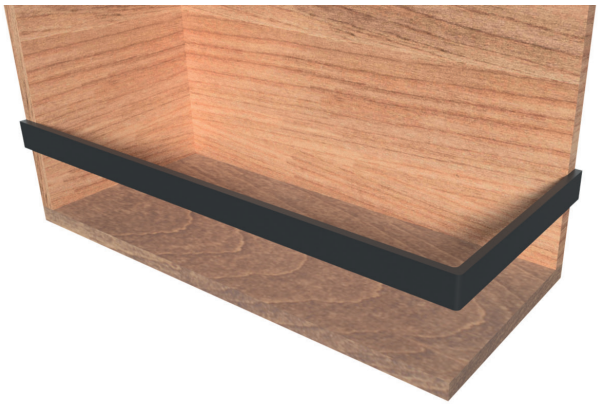
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[Turn over

1. (continued)

(b) Aluminium was used for the rail shown below, as it is a strong and durable material.



(i) State **two other** reasons why aluminium is a suitable material for the rail. **2**

The edges of the aluminium rail were finished before bending.

(ii) Outline **two** stages in finishing the edges of the aluminium rail. **2**

You may use sketches to illustrate your answer in the box below.

1. (b) (continued)

The aluminium rail was formed to a right angle.

(iii) Describe how the right-angle could be formed accurately, with reference to workshop tools. **2**

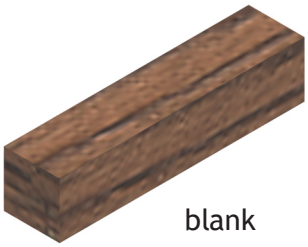
You may use sketches to illustrate your answer in the box below.



[Turn over

1. (continued)

(c) The peg was made from a hardwood blank and turned into the shape shown below.



blank



turned peg

(i) Name a suitable hardwood for the peg. **1**

(ii) Outline **four** stages in preparing the blank before fitting it on the wood lathe, with reference to workshop tools. **4**

You may use sketches to illustrate your answer in the box below.

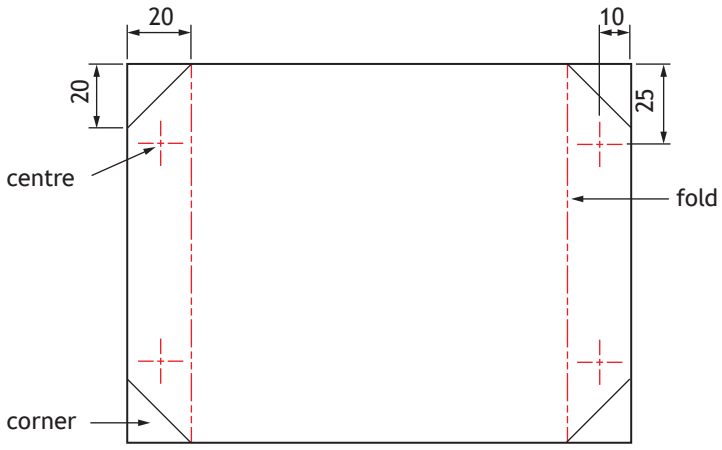
1. (c) (continued)

The peg was finished while on the wood lathe.

(iii) Outline **two** ways of ensuring a high-quality finish on the peg when using the wood lathe. **2**

1. (continued)

(d) The spoon holder was marked out on a mild steel sheet as shown below.



(i) Describe **three** stages in marking out the mild steel sheet metal, with reference to workshop tools. **3**

You may use sketches to illustrate your answer in the box below.

1. (d) (continued)

The corners were removed from the mild steel sheet metal before forming.

(ii) Name a suitable tool that could be used to cut the corners. 1

After forming the curve on the spoon holder, a finish was applied.

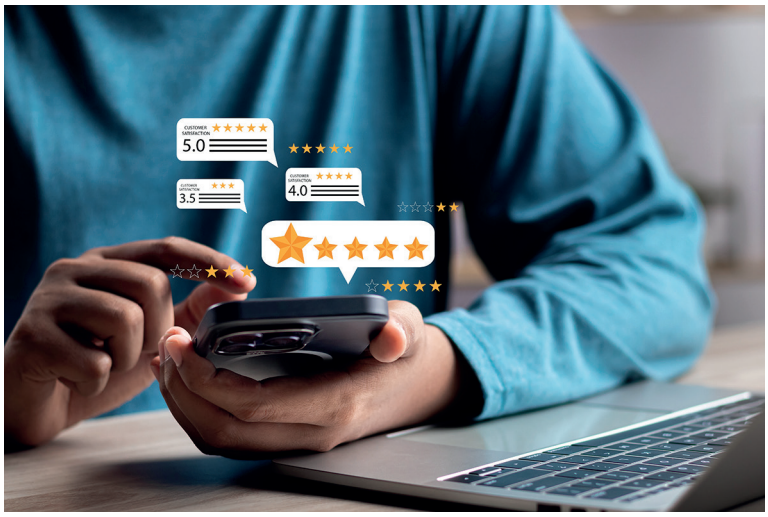


(iii) Name a suitable finish for the spoon holder. 1

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

2. User trips and questionnaires are research techniques often used by designers.



(a) Outline **three** key stages of a user trip. **3**

(b) Outline **three** benefits of using a questionnaire to carry out research. **3**

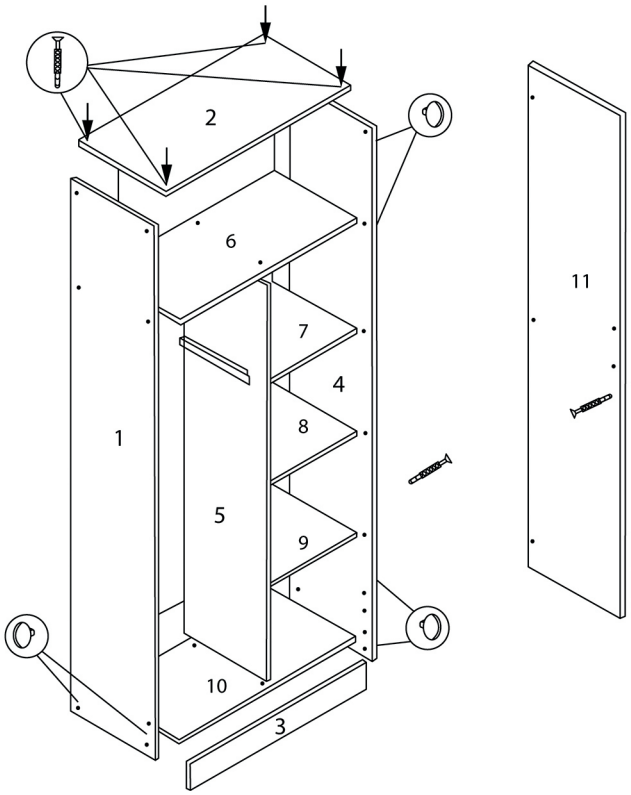
3. Idea generation techniques are often used by designers.

Outline the key stages of morphological analysis. **3**

You may use sketches to illustrate your answer in the box below.

[Turn over

4. Different graphic techniques can be used throughout the design process.
An exploded view is shown below.



You must give different responses in (a) and (b).

(a) Outline **two** pieces of information that can be gained from an exploded view. **2**

4. (continued)

A rendered pictorial is shown below.



You must give different responses in (a) and (b).

(b) Outline **two** reasons for using rendered pictorials in the design process. **2**

5. Models are often used during the design process.

Describe the benefits of using models within the design process. **2**

6. (a) A treadmill is shown below.



You must give different responses for (a)(i), (ii) and (iii).

Describe how the following design factors may have influenced the design of the treadmills:

(i) Ergonomics. **4**

6. (a) (continued)

(ii) Function. **3**

(iii) Safety. **3**

6. (continued)

A water bottle is shown below.



(b) Describe the aesthetic aspects of the water bottle. **3**

6. (continued)

(c) Products related to the fitness industry are often influenced by technology push and market pull.

(i) Describe what is meant by technology push. **1**

(ii) Describe what is meant by market pull. **1**

[Turn over

7. A toy car and packaging are shown below.



SJBright/Shutterstock.com



plastic packaging

Different reasons must be given for the suitability of each material.

(a) Polystyrene was used to make the plastic packaging shown above.

(i) State **two** reasons why polystyrene is a suitable material for the plastic packaging. **2**

A metal alloy was used to make the body of the car.

(ii) State **two** reasons why the metal alloy is a suitable material for the body of the car. **2**

7. (continued)

The toy car was die cast.

(b) State **two** identifying features that show products are die cast. **2**

The plastic packaging was vacuum formed around a pattern.

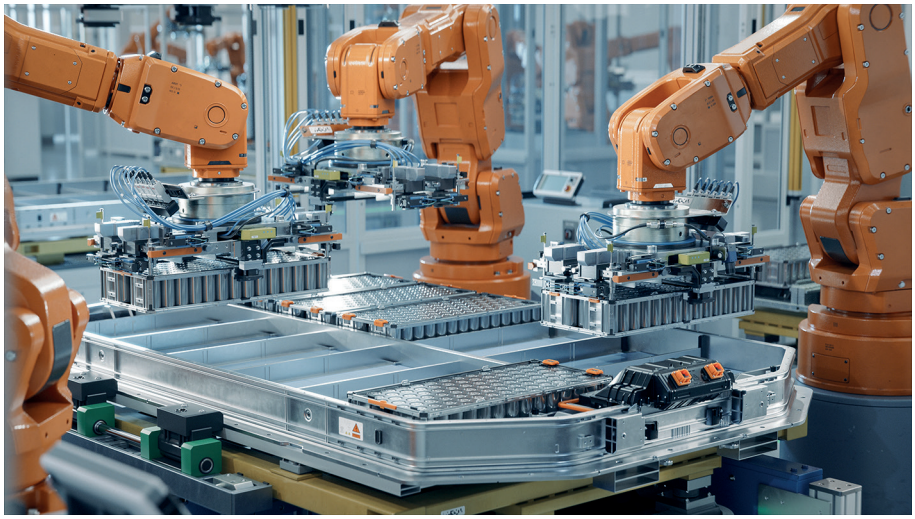
(c) State a manufacturing reason for each feature of vacuum forming:
A different reason must be given for each feature.

(i) Tapered edges. **1**

(ii) Rounded corners. **1**

[Turn over

8. Computer-Aided Manufacture (CAM) is used to help with mass manufacturing products.



You must give different responses in (a) and (b).

(a) Describe the benefits of using CAM to the **manufacturer**. **4**

8. (continued)

(b) Describe the impact of mass manufacture on **society**. **2**

[Turn over

9. Standard components are used in the mass manufacture of products.



Outline **three** benefits of using standard components in products. **3**

10. To reduce their impact on the environment, companies can take steps to increase the life expectancy of products.



Describe the steps that could be taken to increase the life expectancy of products. **3**

[END OF QUESTION PAPER]

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